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Random Sampling of Mellin Band-Limited Signals

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ABSTRACT
In this paper, we address the random sampling problem for the class of functions in the space of Mellin band-limited functions $\mathcal{B}_T$, which are concentrated on a bounded cube. It is established that any Mellin band-limited function can be approximated by an element in a finite-dimensional subspace of $\mathcal{B}_T$. Utilizing the notion of covering number and Bernstein's inequality to the sum of independent random variables, we prove that the probabilistic sampling inequality holds for the set of concentrated signals in $\mathcal{B}_T$ with an overwhelming probability provided the sampling size is large enough.

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1. Introduction

The theory of sampling and reconstruction has been receiving significant attention from the broad areas of approximation theory, signal processing, and digital communication. The sampling problem for the function space V is mainly concerned with the stable recovery of a function $f \in V$ without losing any information from its sample values $\{f(x_j)\}$ on some countable set $\{x_j : j \in J\}$. For the space of Mellin band-limited functions $\mathcal{B}_T$ (defined in Section 2), the sampling problem is equivalent to finding a sampling set $X = \{x_j : j \in J\}$ such that the sampling inequality

$$A\|f\|^2_{X^2_c(\mathbb{R}^n_+)} \leq \sum_{j\in J} |f(x_j)|^2 x_j^{2c} \leq B\|f\|^2_{X^2_c(\mathbb{R}^n_+)}, \tag{1.1}$$

holds for all f in $\mathcal{B}_T$, for some $A, B > 0$. The sample set X which satisfy (1.1) is called a stable sampling set. Here, $\mathbb{R}^n_+$ denotes the set points in the positive quadrant of Euclidean space $\mathbb{R}^n$. For $c \in \mathbb{R}^n$, the Banach space $X^2_c(\mathbb{R}^n_+)$ is a collection of all measurable function f with

$$\|f\|^2_{X^2_c(\mathbb{R}^n_+)} = \int_{\mathbb{R}^n_+} |x^c f(x)|^2\, x^{-1} dx < \infty.$$

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An important result in this direction was collectively given by Bertero and Pike [1], and Gori [2] which asserts that for any $T > 0$, the set of exponentially spaced sample points $\{e^{\frac{k}{T}} : k \in \mathbb{Z}\}$ is a stable sampling set for the space of Mellin band-limited functions defined on $\mathbb{R}_+$ and the reconstruction formula is given by

$$f(x) = \sum_{k\in\mathbb{Z}} f(e^{\frac{k}{T}}) \operatorname{lin}_{\frac{c}{T}}(e^{-k}x^T), \qquad x \in \mathbb{R}_+, \tag{1.2}$$

where $\operatorname{lin}_c(x) = x^{-c}\operatorname{sinc}(\log x)$, $c \in \mathbb{R}$, and $\operatorname{sinc}(u) = \frac{\sin \pi u}{\pi u}$. In particular, we have $A = B = T$. The exponential sampling method is also useful to deal with various real-world problems arising in science and engineering (see [3, 4]), in particular in the application where the signal information is accumulated near zero. Butzer and Jansche [5] pioneered the mathematical study of the exponential sampling formula (1.2) employing tools from the theory of Mellin transform. Bardaro et al. [6] generalized the exponential sampling formula to the functions that may not necessarily be Mellin band-limited. The Mellin theory has been playing a crucial role in the study of exponential sampling formula and its generalizations. Some of the developments in this direction can be observed in [7–11].

Riemann first employed Mellin transform in his renowned memoir on the Riemann zeta function (see [12]) and later, R. G. Mamedov investigated some properties of Mellin transform [13]. Butzer and Jansche [12, 14] presented various theoretical aspects of Mellin transform along with its applications to different areas of mathematical analysis. Some well known results in the context of Fourier band-limited functions were generalized for the space of Mellin band-limited functions, such as the reproducing formula [15], Parseval theorem [16], and the analogue of theory of Fourier series in the Mellin setting [17]. The studies in [18, 19] examined the notion of fractional integration and differentiation in the Mellin setting. In this paper, we have studied the random sampling problem for the space of Mellin band-limited functions.

In recent years, signals have been measured and recovered from their sample values at randomly distributed sample points in the field of image processing [20], compressed sensing [21, 22], and machine learning [23, 24]. Bass and Gröchenig [25] showed that for any random distribution on an unbounded domain, the signal recovery fails almost surely for the space of Fourier band-limited functions. Hence, the random sampling problem deals with the stability of the random sample set for a given class of function space. In this article, we study the random sampling problem for the class of Mellin band-limited functions. In particular, for uniformly distributed random samples $X = \{x_j : j \in J\}$ drawn from bounded subset of $\mathbb{R}^n_+$, we investigate the following probability bound

$$P\Big(A\|f\|^2_{X^2_c(\mathbb{R}^n_+)} \le \sum_{j\in J} |f(x_j)|^2 x_j^{2c} \le B\|f\|^2_{X^2_c(\mathbb{R}^n_+)}\Big) \ge 1 - \epsilon,$$

for some class of Mellin band-limited functions, where $\epsilon > 0$ is arbitrary small.

The random sampling problem was studied for the space of multivariate trigonometric polynomials [26, 27]. In the case of infinite-dimensional space, Bass and Gröchenig first established [25, 28] that the class of Fourier band-limited functions concentrated on a bounded domain $C \subset \mathbb{R}^n$ can be stably recovered from its samples at uniformly distributed random points on C. The random sampling problem had been investigated for the shift-invariant spaces [29–31], the signal space of finite rate of innovation [32], the image space of an idempotent integral operator [33–35]. In recent years, the related random sampling problem has been studied for the spherical harmonic function space [36].

A Mellin band-limited function can not be Fourier band-limited at the same time, and it is analytically extensible to the Riemann surface of the (complex) logarithm (see [37]). Since one must appropriately extend the idea of the Bernstein spaces, involving Riemann surfaces, the theory of Mellin band-limited functions differs greatly from the theory of Fourier band-limited functions. The main aim of the paper is to study the stability of the random sampling set for the space of Mellin band-limited functions. Motivated from [25], we consider the problem of random sampling for the class of δ-concentrated Mellin band-limited functions on $C_R := [1/R, R]^n \subset \mathbb{R}^n_+$, i.e.,

$$\mathcal{B}_{T,\delta} := \left\{ f \in \mathcal{B}_T : \int_{C_R} |f(x)|^2 x^{2c} x^{-1} dx \geq (1-\delta) \|f\|^2_{X^2_c(\mathbb{R}^n_+)} \right\},$$

where $\delta \in (0,1)$, $R > 1$, and $c \in \mathbb{R}^n$.

The paper is organized as follows. Section 2 provides some preliminaries required to derive the desired results. In Section 3, we first establish that any element in $\mathcal{B}_T$ can be approximated by an element in a finite-dimensional subspace of $\mathcal{B}_T$ with the help of the representation formula (1.2). Then we show that any bounded subset of $\mathcal{B}_{T,\delta}$ is totally bounded with respect to $\|\cdot\|_{X^\infty_c(C_R)}$. In the end, by using the notion of covering number and the well-known Bernstein's inequality, we prove that the random sampling inequality holds with overwhelming probability for the functions concentrated on the compact set.

2. Preliminaries

In this section, we define some preliminary results, and basic notations used in the rest of the paper are given in Table 1.

For $1 \leq p < \infty$, let $L^p(\mathbb{R}^n_+)$ be the space of p-integrable functions in the Lebesgue sense on $\mathbb{R}^n_+$ with usual $p-$norm. Moreover, $L^\infty(\mathbb{R}^n_+)$ denotes the class of bounded measurable functions defined on $\mathbb{R}^n_+$ with norm $\|f\|_{L^\infty(\mathbb{R}^n_+)} := ess \sup_{x \in \mathbb{R}^n_+} |f(x)|$. Let $c \in \mathbb{R}^n$ be fixed and for $1 \leq p \leq \infty$, we define the space

$$X^p_c(\mathbb{R}^n_+) := \{ f : \mathbb{R}^n_+ \to \mathbb{C} : (\cdot)^c (\cdot)^{-1/p} f(\cdot) \in L^p(\mathbb{R}^n_+) \}$$

Table 1. Some notations used for the rest of the paper.

Notation	Remark
$x := \big(x(1), x(2), \ldots, x(n)\big) \in \mathbb{R}^n$	$x(i) \in \mathbb{R}$
$\frac{x}{y} := \Big(\frac{x(1)}{y(1)}, \frac{x(2)}{y(2)}, \ldots, \frac{x(n)}{y(n)}\Big)$	$x, y \in \mathbb{R}^n_+$
$\log x := \big(\log x(1), \log x(2), \ldots, \log x(n)\big)$	$x \in \mathbb{R}^n_+$
$x^c := x(1)^{c(1)} \cdot x(2)^{c(2)} \cdots x(n)^{c(n)}$	$x \in \mathbb{R}^n_+, c \in \mathbb{R}^n$
$\alpha^x := \big(\alpha^{x(1)}, \alpha^{x(2)}, \ldots, \alpha^{x(n)}\big)$	$\alpha \in \mathbb{R}_+, x \in \mathbb{R}^n$
$x^\alpha := x(1)^\alpha \cdot x(2)^\alpha \ldots x(n)^\alpha$	$\alpha \in \mathbb{R}, x \in \mathbb{R}^n$
$\operatorname{sinc}(x) := \frac{\sin \pi x(1)}{\pi x(1)} \cdot \frac{\sin \pi x(2)}{\pi x(2)} \cdots \frac{\sin \pi x(n)}{\pi x(n)}$	$x \in \mathbb{R}^n$
$\operatorname{lin}_c(x) := x^{-c} \operatorname{sinc}(\log x)$	$x \in \mathbb{R}^n_+$

equipped with norm

$$\|f\|_{X^p_c(\mathbb{R}^n_+)} = \|(\cdot)^c (\cdot)^{-1/p} f(\cdot)\|_{L^p(\mathbb{R}^n_+)}.$$

It is important to mention that $X^p_c(\mathbb{R}^n_+)$ is a Banach space (see [14]). Moreover, for $p = 2$, $X^2_c(\mathbb{R}^n_+)$ is a Hilbert space with the inner product

$$\langle f, g \rangle_{X^2_c(\mathbb{R}^n_+)} := \int_{\mathbb{R}^n_+} f(x) \overline{g(x)} x^{2c} x^{-1} dx.$$

Let $c + it =: s \in \mathbb{C}^n$. Then the multi-dimensional Mellin transform of any function $f \in X^1_c(\mathbb{R}^n_+)$ is given by (see [38])

$$\hat{M}[f](s) := \int_{\mathbb{R}^n_+} f(x)\, x^s\, x^{-1} dx.$$

For a fixed $c \in \mathbb{R}^n$, the corresponding inverse Mellin transform is defined as

$$\hat{M}^{-1}[f](x) := \frac{1}{(2\pi i)^n} \int_{c+i\mathbb{R}^n} \hat{M}[f](s)\, x^{-s}\, ds.$$

Definition 2.1. For $c, t \in \mathbb{R}^n$ and $T > 0$, any function $f \in X^2_c(\mathbb{R}^n_+)$ is said to be Mellin band-limited to $[-T, T]^n$ if $\hat{M}[f](c + it) = 0$ for all $\|t\|_\infty > T$. We denote $\mathcal{B}_T$ the space of all Mellin band-limited functions, i.e.,

$$\mathcal{B}_T = \Big\{ f \in X^2_c(\mathbb{R}^n_+) : \hat{M}[f](c + it) = 0 \ \text{ for all } \|t\|_\infty > T \Big\}.$$

A Hilbert space H of functions defined on Λ is a reproducing kernel Hilbert space if for each $x \in \Lambda$, the point evaluation functional $g \mapsto g(x)$ is continuous, i.e., for each $x \in \Lambda$, there exists positive constant m_x such that

$$|g(x)| \le m_x \|g\|, \qquad \forall g \in H.$$

Lemma 2.2. *The space $\mathcal{B}_T$ is a reproducing kernel space. Moreover, for $x \in \mathbb{R}^n_+$, we have*

$$|f(x)| \le x^{-c} \|f\|_{X^2_c(\mathbb{R}^n_+)}, \quad \forall f \in \mathcal{B}_T.$$

Proof Let f be Mellin band-limited to $[-T, T]$, and $T > 0$. Then from [15, Theorem 4], we have

$$f(x) = T \int_{\mathbb{R}_+} f(y) \operatorname{lin}_{\frac{c}{T}} \left(\frac{x}{y}\right)^T \frac{dy}{y}, \qquad x \in \mathbb{R}_+. \tag{2.1}$$

Using n-dimensional Mellin transform and inverse Mellin transform, we extend (2.1) for n-dimensional Mellin band-limited function. In particular, we have

$$f(x) = T^n \int_{\mathbb{R}_+^n} f(y) \operatorname{lin}_{\frac{c}{T}} \left(\frac{x}{y}\right)^T y^{-1} dy, \tag{2.2}$$

where $\operatorname{lin}_{\frac{c}{T}} \left(\frac{x}{y}\right)^T := x^{-c} y^c \operatorname{sinc}(T \log x - T \log y)$. In view of (2.2) and Hölder's inequality, we have

$$\begin{aligned}
|f(x)| &\leq T^n x^{-c} \left(\int_{\mathbb{R}_+^n} |f(y)|^2 y^{2c} y^{-1} dy \right)^{1/2} \left(\int_{\mathbb{R}_+^n} \operatorname{sinc}^2 \left(T \log(x/y)\right) y^{-1} dy \right)^{1/2} \\
&\leq x^{-c} \|f\|_{X_c^2(\mathbb{R}_+^n)} \|\operatorname{sinc}\|_{L^2(\mathbb{R}^n)} \\
&\leq x^{-c} \|f\|_{X_c^2(\mathbb{R}_+^n)}.
\end{aligned}$$

This completes the proof. □

Now, we see an example of a Mellin band-limited function, which is important due to its major application in approximation theory. For instance, it plays a vital role in the study of the convergence of generalized exponential sampling series, see [6].

Example 2.3. [6] Let $c \in \mathbb{R}, \alpha \geq 1$, and $k \in \mathbb{N}$. Then the *Mellin Jackson kernel* is defined as

$$J_{\alpha,k}(x) := C_{\alpha,k}\, x^{-c} \operatorname{sinc}^{2k} \left(\frac{\log x}{2\alpha k \pi} \right),$$

where $x \in \mathbb{R}_+$ and $C_{\alpha,k}^{-1} := \displaystyle\int_0^\infty \operatorname{sinc}^{2k} \left(\frac{\log x}{2\alpha k\pi} \right) \frac{dx}{x}$. Since $\hat{M}[J_{\alpha,k}](c + it) = 0$ for $|t| \geq \frac{1}{\alpha}$, the function $J_{\alpha,k}$ is Mellin band-limited. In particular, for $k = 1$, it is reduced to the well-known kernel function referred to as the *Mellin Fejer kernel*, which is given by

$$F_\rho^c(x) = \frac{\rho}{2\pi} x^{-c} \operatorname{sinc}^2 \left(\frac{\rho \log \sqrt{x}}{\pi} \right).$$

It is easy to verify that $J_{\alpha,k}$ is not in $L^p(\mathbb{R}_+)$ for $1 \leq p < \infty$, hence Fourier transform of $J_{\alpha,k}$ does not exists.

3. Main results

In this section, we discuss the proposed results of the paper. We see that for each $f \in \mathcal{B}_T$, the following representation holds:

$$f(x) = \sum_{k\in\mathbb{Z}^n} f(e^{k/T}) \operatorname{lin}_{\frac{c}{T}}(e^{-k}x^T), \qquad x \in \mathbb{R}^n_+.$$

Now we aim to show that any $f \in \mathcal{B}_T$ can be approximated on C_R by an element from a finite-dimensional subspace $\mathcal{B}_T^N$ of $\mathcal{B}_T$ where $\mathcal{B}_T^N$ is defined by

$$\mathcal{B}_T^N = \Bigg\{ \sum_{k\in\mathbb{Z}^n\cap[-\frac{N}{2},\frac{N}{2}]^n} c_k \operatorname{lin}_{\frac{c}{T}}\left(e^{-k}(\cdot)^T\right) : c_k \in \mathbb{R}\Bigg\}.$$

Theorem 3.1. *Let $\epsilon > 0$. Then for each $f \in \mathcal{B}_T$, there exists $f_N \in \mathcal{B}_T^N$ such that*

$$\|f - f_N\|_{X_c^\infty(C_R)} < \epsilon\|f\|_{X_c^2(\mathbb{R}^n_+)},$$

whenever $N > 4T\pi^{-2}\epsilon^{-2/n} + 2T\log R$.

Proof For any $f \in \mathcal{B}_T$ and $x \in C_R$, we have $f_N \in \mathcal{B}_T^N$ as

$$f_N(x) = \sum_{k\in\mathbb{Z}^n\cap[-\frac{N}{2},\frac{N}{2}]^n} f(e^{k/T}) \operatorname{lin}_{\frac{c}{T}}(e^{-k}x^T).$$

In view of Cauchy-Schwarz inequality, we obtain

$$\begin{aligned}
&x^c\left|f(x) - f_N(x)\right| \\
&= \left|x^c \sum_{k\in\mathbb{Z}^n\setminus[-\frac{N}{2},\frac{N}{2}]^n} f(e^{k/T}) \operatorname{lin}_{\frac{c}{T}}(e^{-k}x^T)\right| \\
&= \left| \sum_{k\in\mathbb{Z}^n\setminus[-\frac{N}{2},\frac{N}{2}]^n} f(e^{k/T})\, e^{\frac{\langle c,k\rangle_2}{T}} \operatorname{sinc}(T\log x - k)\right| \\
&\le \left(\sum_{k\in\mathbb{Z}^n\setminus[-\frac{N}{2},\frac{N}{2}]^n} |f(e^{k/T})|^2 e^{\frac{2\langle c,k\rangle_2}{T}} \right)^{\frac{1}{2}} \\
&\quad \left(\sum_{k\in\mathbb{Z}^n\setminus[-\frac{N}{2},\frac{N}{2}]^n} \operatorname{sinc}^2(T\log x - k) \right)^{\frac{1}{2}}.
\end{aligned} \tag{3.1}$$

In view of [15, Theorem 4], for $j, k \in \mathbb{Z}^n$, we have

$$T^n e^{\frac{-2\langle c,k\rangle_2}{T}} \langle \operatorname{lin}_{\frac{c}{T}}(e^{-k}x^T), \operatorname{lin}_{\frac{c}{T}}(e^{-j}x^T)\rangle_{X_c^2(\mathbb{R}^n_+)} = \operatorname{lin}_{\frac{c}{T}}(e^{j-k}) = \delta_{j,k}\,,$$

where $\delta_{j,k} = \begin{cases} 1, & \text{if } j = k, \\ 0, & \text{if } j \neq k. \end{cases}$

This gives

$$\begin{aligned}\|f\|^2_{X^2_c(\mathbb{R}^n_+)} &= \sum_{k\in\mathbb{Z}^n}\sum_{j\in\mathbb{Z}^n} f(e^{k/T})\overline{f(e^{j/T})}\left\langle \operatorname{lin}_{\frac{c}{T}}(e^{-k}x^T), \operatorname{lin}_{\frac{c}{T}}(e^{-j}x^T)\right\rangle_{X^2_c(\mathbb{R}^n_+)} \\ &= \frac{1}{T^n}\sum_{k\in\mathbb{Z}^n} |f(e^{k/T})|^2 e^{\frac{2\langle c,k\rangle_2}{T}}.\end{aligned}$$

Now for $x \in C_R$, we have

$$\begin{aligned}\sum_{k\in\mathbb{Z}^n\setminus[-\frac{N}{2},\frac{N}{2}]^n} \operatorname{sinc}^2(T\log x - k) &\le \pi^{-2n}\sum_{k\in\mathbb{Z}^n\setminus[-\frac{N}{2},\frac{N}{2}]^n}\left(\prod_{i=1}^n \frac{\sin^2\pi(T\log x_i - k_i)}{(T\log x_i - k_i)^2}\right) \\ &\le \pi^{-2n}\int_{\mathbb{R}^n\setminus[-\frac{N}{2},\frac{N}{2}]^n}\left(\prod_{i=1}^n \frac{1}{(T\log x_i - y_i)^2}\right)dy \\ &\le \pi^{-2n}\int_{\mathbb{R}^n\setminus[-\frac{N}{2}+T\log R,\frac{N}{2}-T\log R]^n} y^{-2}\,dy \\ &= \frac{4^n}{\pi^{2n}(N-2T\log R)^n}.\end{aligned}$$

On combining these estimates, we obtain from (3.1) that

$$\|f - f_N\|_{X^\infty_c(C_R)} \le \left(T^{n/2}\|f\|_{X^2_c(\mathbb{R}^n_+)}\right)\left(\frac{4}{\pi^2(N-2T\log R)}\right)^{n/2}.$$

This implies that if $N > 4T\pi^{-2}\epsilon^{-2/n} + 2T\log R$, then

$$\|f - f_N\|_{X^\infty_c(C_R)} < \epsilon\|f\|_{X^2_c(\mathbb{R}^n_+)}.$$

This completes the proof. □

In order to derive the desired probability estimate for the sampling inequality (1.1) to hold, we first prove that the bounded subset of δ-concentrated functions is totally bounded.

Lemma 3.2. *The set $\mathcal{B}^*_{T,\delta} := \left\{f \in \mathcal{B}_{T,\delta} : \|f\|_{X^2_c(\mathbb{R}^n_+)} = 1\right\}$ is totally bounded with respect to $\|\cdot\|_{X^\infty_c(C_R)}$.*

Proof It follows similar arguments as in [34, Lemma 2.4]. By Theorem 3.1, for given $\epsilon > 0$ and $f \in \mathcal{B}^*_{T,\delta}$, there exists $f_N \in \mathcal{B}^N_T$ such that $\|f - f_N\|_{X^\infty_c(C_R)} < \frac{\epsilon}{2}$. Let $\overline{B(0;\epsilon)}$ represents the closed ball in $\mathcal{B}^N_T$ having radius ϵ and centred at origin. Now from Lemma 2.2, we have $\|f\|_{X^\infty_c(C_R)} \le 1$ which implies that $f_N \in \overline{B\left(0; 1+\frac{\epsilon}{2}\right)}$. Since $\overline{B\left(0; 1+\frac{\epsilon}{2}\right)}$ is totally bounded, the desired result follows. □

Definition 3.3. Let $(X, \|\cdot\|)$ be a Banach space. For $\beta > 0$, The covering number of a compact subset K of X is defined by

$$\mathcal{N}(K, \beta) := \min \left\{ \ell \in \mathbb{N} : \exists\, x_1, x_2, \ldots, x_\ell \in X \text{ such that } K \subset \bigcup_{m=1}^{\ell} B(x_m; \beta) \right\},$$

where $B(x_m; \beta)$ represents open ball in X of radius β centered at x_m.

The following lemma provides a bound for the covering number of any closed ball in a finite-dimensional Banach space.

Lemma 3.4. [24] *Let X be a Banach space of dimension d and $\overline{B(0; s)}$ represents the closed ball of radius s centered at the origin. Then the minimum number of open balls of radius r required to cover $\overline{B(0; s)}$ is bounded by $\left(\frac{4s}{r}\right)^d$.*

Remark 3.5. From Theorem 3.1 and Lemma 3.2, for $f \in \mathcal{B}^*_{T,\delta}$, it can be seen that for $\|f - f_N\|_{X_c^\infty(C_R)} < \frac{\epsilon}{2}$, we need to choose N in such a way that

$$N > \left\lceil 4T\pi^{-2} 2^{\frac{n}{2}} \epsilon^{-2/n} \right\rceil + 2T \log R.$$

Particularly, if we set $N = 4T\pi^{-2} 2^{\frac{n}{2}} \epsilon^{-2/n} + 2T \log R + 1$, then the dimension of B_N^T will be bounded by

$$\begin{aligned} N^n &= \left[T 2^{\frac{n}{2}+2} \pi^{-2} \epsilon^{-2/n} + 2T \log R + 1 \right]^n \\ &\le 2^n \left[\frac{T^n 2^{n\left(\frac{n}{2}+2\right)} \pi^{-2n}}{\epsilon^2} + (2T \log R + 1)^n \right] := d_\epsilon. \end{aligned}$$

Now from Lemmas 3.2 and 3.4, the bound for the covering number of $\mathcal{B}^*_{T,\delta}$ is given by

$$\mathcal{N}\left(\mathcal{B}^*_{T,\delta}, \epsilon\right) \le \mathcal{N}\left(\overline{B\left(0; 1 + \frac{\epsilon}{2}\right)}, \frac{\epsilon}{2} \right) \le \exp\left(d_\epsilon \, \log\left(\frac{16}{\epsilon}\right)\right). \tag{3.2}$$

3.1. Random sampling

In this section, we deduce the probability estimate concerning the random sampling inequality (3.5). We first define independent random variables on $\mathcal{B}_{T,\delta}$ by using random samples. The main tool to derive the probability estimate is the Bernstein inequality, which utilizes the upper bounds of the random variables and their variance.

Lemma 3.6 (Bernstein's Inequality [39]). *Let Y_j, $j = 1, 2, \ldots, r$ be a sequence of bounded, independent random variables with $E[Y_j] = 0$, $Var[Y_j] \le \sigma^2$, and*

$\|Y_j\|_\infty \le M$ *for* $j = 1, 2, \ldots, r$. *Then*

$$P\Big(\Big|\sum_{j=1}^{r} Y_j\Big| \ge \lambda\Big) \le 2\exp\Big(-\frac{\lambda^2}{2r\sigma^2 + \frac{2}{3}M\lambda}\Big). \tag{3.3}$$

Let $S := \{x_j : j \in \mathbb{N}\}$ be i.i.d. random variables distributed uniformly on cube C_R. For $f \in \mathcal{B}_{T,\delta}$, we define random variable as follows:

$$Z_j(f) = |f(x_j)|^2 x_j^{2c} x_j^{-1} - \frac{R^n}{(R^2-1)^n}\int_{C_R} |f(x)|^2 x^{2c} x^{-1} dx. \tag{3.4}$$

The set $\{Z_j(f)\}_{j\in\mathbb{N}}$ is a sequence of independent random variables. We also see that

$$E\big[Z_j(f)\big] = \frac{R^n}{(R^2-1)^n}\int_{C_R}\Big[|f(y)|^2 y^{2c} y^{-1} - \frac{R^n}{(R^2-1)^n}\int_{C_R}|f(x)|^2 x^{2c} x^{-1} dx\Big] dy = 0.$$

Lemma 3.7. *Let* $f, g \in \mathcal{B}_T$ *with* $\|f\|_{X_c^2(\mathbb{R}_+^n)} = \|g\|_{X_c^2(\mathbb{R}_+^n)} = 1$. *Then the following estimates hold:*

(i) $Var\big[Z_j(f)\big] \le \frac{R^{2n}}{(R^2-1)^n}$,
(ii) $\|Z_j(f)\|_\infty \le R^n$,
(iii) $\|Z_j(f) - Z_j(g)\|_\infty \le 2R^n \|f - g\|_{X_c^\infty(C_R)}$.

Proof

(i) Using the definition of variance and (3.4), we obtain

$$\begin{aligned}
Var\big[Z_j(f)\big] &= E\big[|f(x_j)|^4 x_j^{4c} x_j^{-2}\big] - \big[E\big(|f(x_j)|^2 x_j^{2c} x_j^{-1}\big)\big]^2 \\
&\le E\big[|f(x_j)|^4 x_j^{4c} x_j^{-2}\big] \\
&= \frac{R^n}{(R^2-1)^n}\int_{C_R} |f(x)|^4 x_j^{4c} x_j^{-2} dx \\
&\le \frac{R^{2n}}{(R^2-1)^n}\|f\|^2_{X_c^\infty(C_R)}\int_{C_R}|f(x)|^2 x_j^{2c} x_j^{-1} dx \\
&= \frac{R^{2n}}{(R^2-1)^n}\|f\|^2_{X_c^\infty(C_R)}\|f\|^2_{X_c^2(C_R)}.
\end{aligned}$$

Since $\|f\|^2_{X_c^\infty(C_R)} \le \|f\|^2_{X_c^2(\mathbb{R}_+^n)}$, and $\|f\|_{X_c^2(\mathbb{R}_+^n)} = 1$, we have

$$Var\big[Z_j(f)\big] \le \frac{R^{2n}}{(R^2-1)^n}.$$

(ii) Now since $f \in \mathcal{B}_T$ and $\|f\|_{X_c^2(\mathbb{R}_+^n)} = 1$, we can write

$$\begin{aligned}\|Z_j(f)\|_\infty &= \sup_{x_j \in S} \left| |f(x_j)|^2 x_j^{2c} x_j^{-1} - \frac{R^n}{(R^2-1)^n} \int_{C_R} |f(x)|^2 x_j^{2c} x_j^{-1} dx \right| \\ &\leq \max\left\{ R^n \|f\|^2_{X_c^\infty(C_R)}, \frac{R^n}{(R^2-1)^n} \int_{C_R} |f(x)|^2 x_j^{2c} x_j^{-1} dx \right\} \\ &= R^n \|f\|^2_{X_c^\infty(C_R)} \leq R^n.\end{aligned}$$

(iii) Similar argument as in (ii), we have

$$\begin{aligned}&\|Z_j(f) - Z_j(g)\|_\infty \\ &= \sup_{x_j \in S} \Big| \left(|f(x_j)|^2 - |g(x_j)|^2\right) x_j^{2c} x_j^{-1} \\ &\quad - \frac{R^n}{(R^2-1)^n} \int_{C_R} (|f(x)|^2 - |g(x)|^2)\, x_j^{2c} x_j^{-1} dx \Big| \\ &\leq \max\Big\{ (|f(x)| - |g(x)|)(\,|f(x)| + |g(x)|) x^{2c} x^{-1}, \\ &\qquad\qquad \frac{R^n}{(R^2-1)^n} \int_{C_R} (|f(x)|^2 - |g(x)|^2) x^{2c} x^{-1} dx \Big\} \\ &\leq 2R^n \|f - g\|_{X_c^\infty(C_R)}.\end{aligned}$$

This completes the proof. □

Using these estimates, we now proceed toward probability estimation. The following result will be helpful in this direction.

Theorem 3.8. *Let $\{x_j : j \in \mathbb{N}\}$ be the i.i.d. random variables drawn uniformly from the cube C_R. Then the following holds*

$$\begin{aligned}&P\left(\sup_{f \in \mathcal{B}^*_{T,\delta}} \left| \frac{1}{r} \sum_{j=1}^{r} Z_j(f) \right| \geq \epsilon \right) \\ &\quad \leq 2 \exp\Bigg[2^n \left(\frac{16T\pi^{-2} R^{2n} 2^{n(\frac{n}{2}+2)}}{\epsilon^2} + (2T \log R + 1)^n \right) \\ &\qquad\qquad \log\left(\frac{64R^n}{\epsilon}\right) - \frac{3r\epsilon^2 (R^2-1)^n}{4R^n(6R^n + \epsilon(R^2-1)^n)} \Bigg].\end{aligned}$$

Proof Let $M := \mathcal{N}\left(\mathcal{B}^*_{T,\delta}, \frac{\epsilon}{4R^n}\right)$ be the covering number for the set $\mathcal{B}^*_{T,\delta}$ and $f_1, f_2, \ldots, f_M$ be the elements of $\mathcal{B}^*_{T,\delta}$. Then for any $f \in \mathcal{B}^*_{T,\delta}$, there exists f_m,

$1 \le m \le M$ such that $\|f - f_m\|_{X_c^\infty(C_R)} < \dfrac{\epsilon}{4R^n}$. From (iii) of Lemma 3.7, we have

$$\left|\frac{1}{r}\sum_{j=1}^{r}\big(Z_j(f) - Z_j(f_m)\big)\right| \le 2R^n\|f - f_m\|_{X_c^\infty(C_R)} \le \frac{\epsilon}{2}.$$

In view of (3.3), for given f_m, $1 \le m \le M$, we have

$$P\left(\left|\frac{1}{r}\sum_{j=1}^{r} Z_j(f_m)\right| \ge \frac{\epsilon}{2}\right) \le 2\exp\left(-\frac{3r\epsilon^2(R^2-1)^n}{4R^n(6R^n + \epsilon(R^2-1)^n)}\right).$$

For fixed m, we have

$$\begin{aligned}
&P\left(\sup_{\left\{f:\|f-f_m\|_{X_c^\infty(C_R)} \le \frac{\epsilon}{4R^n}\right\}}\left|\frac{1}{r}\sum_{j=1}^{r} Z_j(f)\right| \ge \epsilon\right)\\
&\quad\le P\left(\left|\frac{1}{r}\sum_{j=1}^{r} Z_j(f_m)\right| \ge \frac{\epsilon}{2}\right)\\
&\quad\le 2\exp\left(-\frac{3r\epsilon^2(R^2-1)^n}{4R^n(6R^n + \epsilon(R^2-1)^n)}\right).
\end{aligned}$$

Since M is the covering number for $\mathcal{B}^*_{T,\delta}$, i.e.,

$$\mathcal{B}^*_{T,\delta} \subset \bigcup_{m=1}^{M}\left\{f : \|f - f_m\|_{X_c^\infty(C_R)} \le \frac{\epsilon}{4R^n}\right\},$$

we obtain

$$\begin{aligned}
&P\left(\sup_{f\in\mathcal{B}^*_{T,\delta}}\left|\frac{1}{r}\sum_{j=1}^{r} Z_j(f)\right| \ge \epsilon\right)\\
&\quad\le \sum_{m=1}^{M} P\left(\sup_{\left\{f:\|f-f_m\|_{X_c^\infty(C_R)} \le \frac{\epsilon}{4R^n}\right\}}\left|\frac{1}{r}\sum_{j=1}^{r} Z_j(f)\right| \ge \epsilon\right)\\
&\quad\le 2M\exp\left(-\frac{3r\epsilon^2(R^2-1)^n}{4R^n(6R^n + \epsilon(R^2-1)^n)}\right).
\end{aligned}$$

Using (3.2), we finally get

$$\begin{aligned}
&P\left(\sup_{f\in\mathcal{B}^*_{T,\delta}}\left|\frac{1}{r}\sum_{j=1}^{r} Z_j(f)\right| \ge \epsilon\right)\\
&\quad\le 2\exp\left[2^n\left(\frac{16T^n\pi^{-2n}R^{2n}2^{n\left(\frac{n}{2}+2\right)}}{\epsilon^2} + (2T\log R + 1)^n\right)\right.\\
&\qquad\qquad\left.\log\left(\frac{64R^n}{\epsilon}\right) - \frac{3r\epsilon^2(R^2-1)^n}{4R^n(6R^n + \epsilon(R^2-1)^n)}\right].
\end{aligned}$$

□

We now prove the following main result of the paper.

Theorem 3.9. *Let $\{x_j : j \in \mathbb{N}\}$ be the independent and identically distributed random variables that are uniformly distributed over C_R. For $0 < \mu < 1-\delta$, the following inequality*

$$\frac{(1-\delta-\mu)}{(R^2-1)^n}\|f\|^2_{X_c^2(\mathbb{R}^n_+)} \leq \frac{1}{r}\sum_{j=1}^{r}|f(x_j)|^2 x_j^{2c} \leq \frac{R^{2n}(1+\mu)}{(R^2-1)^n}\|f\|^2_{X_c^2(\mathbb{R}^n_+)}, \tag{3.5}$$

holds for every $f \in \mathcal{B}_{T,\delta}$ with probability at least $1-2\beta\exp\left(-\frac{3r\mu^2}{4(R^2-1)^n(6+\mu)}\right)$, where

$$\beta := \exp\left(2^n\left(\frac{16T^n\pi^{-2n}(R^2-1)^{2n}\,2^{n(\frac{n}{2}+2)}}{\mu^2} + (2T\log R+1)^n\right)\log\left(\frac{64(R^2-1)^n}{\mu}\right)\right).$$

Proof Let $\{x_j : j \in J\}$ be the uniformly distributed samples taken from C_R and define the following event

$$\mathcal{E} = \left\{\sup_{f\in\mathcal{B}^*_{T,\delta}}\left|\frac{1}{r}\sum_{j=1}^{r}Z_j(f)\right| \leq \frac{\mu R^n}{(R^2-1)^n}\right\}.$$

The event $\mathcal{E}$ is equivalent to the event

$$\left|\sum_{j=1}^{r}|f(x_j)|^2x_j^{2c}x_j^{-1} - \frac{rR^n}{(R^2-1)^n}\int_{C_R}|f(x)|^2x_j^{2c}x_j^{-1}\,dx\right| \leq \frac{r\mu R^n}{(R^2-1)^n}, \quad f \in \mathcal{B}^*_{T,\delta}.$$

This implies that for all $f \in \mathcal{B}^*_{T,\delta}$,

$$\frac{rR^n}{(R^2-1)^n}\int_{C_R}|f(x)|^2x^{2c}x^{-1}\,dx - \frac{r\mu R^n}{(R^2-1)^n} \leq \sum_{j=1}^{r}|f(x_j)|^2x_j^{2c}x_j^{-1} \leq \frac{rR^n}{(R^2-1)^n}\int_{C_R}|f(x)|^2x^{2c}x^{-1}\,dx + \frac{r\mu R^n}{(R^2-1)^n}.$$

This gives

$$\frac{(1-\delta-\mu)}{(R^2-1)^n} \leq \frac{1}{r}\sum_{j=1}^{r}|f(x_j)|^2x_j^{2c} \leq \frac{R^{2n}}{(R^2-1)^n}(1+\mu).$$

It can be observed that the event $\mathcal{E}$ leads to the desired sampling inequality (3.5). From Theorem 3.8, the probability that the random samples $\{x_j : j = 1, 2, \dots, r\}$ satisfy the above sampling inequality, is given by

$$\begin{aligned} P(\mathcal{E}) &= 1 - P(\mathcal{E}^c) \\ &\geq 1 - 2\exp\left(-r\alpha + \beta\right), \end{aligned}$$

where $\alpha := \dfrac{3\mu^2}{4(R^2-1)^n(6+\mu)}$, and

$$\beta := 2^n \left(\frac{16\pi^{-2n} T^n (R^2-1)^{2n}\, 2^{n\left(\frac{n}{2}+2\right)}}{\mu^2} + (2T\log R + 1)^n \right) \log\left(\frac{64(R^2-1)^n}{\mu} \right).$$

□

Remark 3.10. From the above estimate of $P(\mathcal{E})$, it is easy to see that the probability approaches to 1 for sufficiently large sample size r. Moreover, the sampling inequality (3.5) is true for large value of r such that $r > \frac{\beta}{\alpha}$.

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